

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 3**

**COURSE CODE: CSA1442 COURSE NAME: Compiler Design for High-Level Programming**

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**COURSE OUTCOME COVERED**

**CO3:** *Analyze syntax-directed translation method using synthesized and inherited attributes to generate a Parser Tree. (BL4)*

*Assessment Tool Weightage: 10%*

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**QUESTIONS: 60 Total Marks:60**

Annexure A

**MCQ**

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**Code of Conduct:**

*I Monika P/192425059 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student: Monika P**

| Criteria                  | Weightage | Level 4 – Exemplary (50–60 Marks) | Level 3 – Proficient (40–49 Marks) | Level 2 – Developing (30–39 Marks) | Level 1 – Beginning (0–29 Marks) |
|---------------------------|-----------|-----------------------------------|------------------------------------|------------------------------------|----------------------------------|
| Number of Correct Answers | 100%      | 50 to 60 correct answers          | 40 to 49 correct answers           | 30 to 39 correct answers           | 0 to 29 correct answers          |
| Accuracy                  | 100%      | Excellent (90–100%)               | Good (75–89%)                      | Average (50–74%)                   | Poor (Below 50%)                 |

Course Faculty

Annexure A

**MCQ**

1. A compiler designer wants to attach semantic meaning to grammar productions in order to generate intermediate representations efficiently. Which compiler concept is most suitable for this purpose?

- A) Lexical buffering
- B) Associating semantic rules with grammar productions
- C) Register allocation
- D) Peephole optimization

**Option: B**

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2. While compiling an arithmetic expression, a compiler internally creates a hierarchical representation showing operators and operands. Which structure is being generated?

- A) Stack frame
- B) Queue structure
- C) Syntax Tree
- D) Linked List

**Option: C**

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3. A compiler evaluates attributes from child nodes before computing the parent node value during semantic analysis. Which type of attribute evaluation is taking place?

- A) Parent-driven evaluation
- B) Synthesized attribute evaluation
- C) Inherited attribute evaluation
- D) Dynamic attribute evaluation

**Option: B**

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4. A language processor uses LR parsing to evaluate semantic information during reductions. Which attribute grammar is naturally suitable in this case?

- A) Context-free attribute grammar
- B) Recursive attribute grammar
- C) S-attributed definition
- D) Circular attribute grammar

**Option: C**

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5. During semantic rule evaluation, a parser requires both parent information and child information while traversing grammar symbols from left to right. Which definition supports this requirement?

- A) S-attributed definition
- B) Static semantic definition
- C) L-attributed definition
- D) Ambiguous definition

**Option: C**

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6. A compiler engineer wants child nodes to be processed before the parent node during semantic evaluation of expressions. Which traversal strategy should be selected?

- A) Preorder traversal
- B) Inorder traversal
- C) Postorder traversal
- D) Level-order traversal

**Option: C**

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7. An expression such as  $(a+b)*c$  is represented internally without unnecessary grammar details. What does this representation mainly capture?

- A) Memory layout
- B) Hierarchical program structure
- C) Register allocation
- D) CPU execution order

**Option: B**

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8. While evaluating semantic rules, information is passed from a parent node toward its child nodes. Which type of attribute is involved?

- A) Synthesized attribute
- B) Terminal attribute
- C) Inherited attribute
- D) Dynamic attribute

**Option: C**

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9. A compiler rejects the statement `int x = "hello";` before execution begins. Which feature of the programming language is responsible for detecting this issue?

- A) Token generation
- B) Type system
- C) Syntax optimization
- D) Loop unrolling

**Option: B**

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10. Consider the statement `float value = 25;`. Which compiler activity occurs automatically here?

- A) Explicit casting
- B) Dynamic conversion
- C) Implicit type conversion
- D) Symbol replacement

**Option: C**

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11. A semantic analyzer verifies whether variables are declared and operations are type-compatible. Which compiler phase performs these tasks?

- A) Lexical Analysis
- B) Syntax Analysis
- C) Semantic Analysis
- D) Linking

**Option: C**

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12. A compiler developer chooses a representation that eliminates unnecessary grammar symbols while preserving expression meaning. Which structure is preferred over a parse tree?

- A) Symbol Table
- B) Syntax Tree
- C) Token Stream
- D) Finite Automaton

**Option: B**

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13. During compilation, all information regarding identifiers, scope, and datatype is maintained centrally. Which compiler component performs this responsibility?

- A) Code Generator
- B) Symbol Table
- C) Syntax Parser
- D) Optimizer

**Option: B**

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14. A compiler uses only synthesized information for semantic evaluation of grammar productions. Which type of attribute grammar is being implemented?

- A) Dynamic grammar
- B) L-attributed grammar
- C) S-attributed grammar
- D) Context-sensitive grammar

**Option: C**

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15. A recursive descent parser performs semantic translation while expanding productions from the start symbol downward. Which translation approach is being followed?

- A) Bottom-up translation
- B) Top-down translation
- C) Shift-reduce translation
- D) Operator precedence translation

**Option: B**

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16. During shift-reduce parsing, intermediate grammar symbols are temporarily stored before reductions occur. Which data structure supports this operation most effectively?

- A) Queue
- B) Heap
- C) Stack
- D) Linked List

**Option: C**

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17. A compiler reports an error when a boolean value is added to an integer variable. What compiler mechanism detected this problem?

- A) Syntax checking
- B) Type checking
- C) Token validation
- D) Memory management

**Option: B**

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18. A programmer declares a variable to store whole numbers without decimal values. Which datatype is most appropriate?

- A) Boolean
- B) Integer
- C) Character
- D) String

**Option: B**

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19. Two array types are considered equivalent because they possess identical internal structures even though their names differ. Which type equivalence principle is applied?

- A) Name equivalence
- B) Structural equivalence
- C) Functional equivalence
- D) Dynamic equivalence

**Option: B**

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20. A compiler implementing LR parsing evaluates semantic rules during reductions. Which evaluation strategy is best suited for this implementation?

- A) Top-down evaluation
- B) Parallel evaluation
- C) Bottom-up evaluation
- D) Recursive evaluation

**Option: C**

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21. During expression analysis, a compiler stores operators as internal nodes and operands as leaves. Which representation is being used?

- A) AVL Tree
- B) Heap Tree
- C) Syntax Tree
- D) B-Tree

**Option: C**

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22. A semantic rule requires information from symbols appearing to the left side of a production. Which attribute grammar supports such evaluation?

- A) S-attributed grammar
- B) L-attributed grammar
- C) Ambiguous grammar
- D) Recursive grammar

**Option: B**

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23. After semantic analysis, a compiler converts validated source constructs into machine-independent instructions. Which compiler phase performs this task?

- A) Lexical Analysis
- B) Intermediate Code Generation
- C) Linking
- D) Parsing

**Option: B**

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24. A programmer intentionally converts a floating-point value into an integer using a language construct. What is this process called?

- A) Promotion
- B) Reduction
- C) Casting
- D) Coercion

**Option: C**

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25. While scanning source code, a compiler groups characters into meaningful units such as keywords and identifiers. Which phase performs this operation?

- A) Semantic Analysis
- B) Syntax Analysis
- C) Lexical Analysis
- D) Optimization



**Option: C**

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26. A parse tree generated for a program statement explicitly contains all grammar symbols used in derivation. What is the primary characteristic of this structure?

- A) Contains machine instructions
- B) Represents grammar symbols
- C) Removes terminals
- D) Stores registers only

**Option: B**

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27. A bottom-up parser starts constructing derivations beginning with identifiers and constants before reaching the start symbol. Which nodes are processed first?

- A) Root nodes
- B) Internal nodes
- C) Leaf nodes
- D) Header nodes

**Option: C**

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28. During assignment checking, a compiler ensures the datatype of the expression matches the datatype of the variable receiving it. Which component performs this verification?

- A) Loader
- B) Semantic checker
- C) Tokenizer
- D) Syntax formatter

**Option: B**

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29. Two variables are treated as equivalent only because they were declared using the same type name. Which equivalence mechanism is being followed?

- A) Structural equivalence
- B) Dynamic equivalence
- C) Name equivalence
- D) Functional equivalence

**Option: C**

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30. A semantic evaluator computes all child expressions before processing the parent expression. Which traversal technique reflects this behavior?

- A) Preorder traversal
- B) Inorder traversal
- C) Postorder traversal
- D) Reverse traversal

**Option: C**

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31. A compiler designer inserts executable semantic instructions directly inside grammar productions to perform translation tasks. What is this mechanism called?

- A) Syntax-directed translation scheme
- B) Token generation mechanism
- C) Recursive grammar scheme
- D) DFA representation

**Option: A**

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32. A parser repeatedly performs shift and reduce operations until the start symbol is produced. Which parsing strategy is being used?

- A) Predictive parsing
- B) Top-down parsing
- C) Bottom-up parsing
- D) Recursive parsing

**Option: C**

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33. A strongly typed programming language prevents invalid arithmetic operations between incompatible datatypes. Which language feature enforces this?

- A) Type system
- B) Code optimizer
- C) Symbol allocator
- D) Loader

**Option: A**

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34. A programmer creates a collection capable of storing multiple values of the same datatype under one name. Which datatype category does this belong to?

- A) Primitive datatype
- B) Derived datatype
- C) Boolean datatype
- D) Character datatype

**Option: B**

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35. In a syntax tree representing an assignment statement, the highest node corresponds to the complete statement meaning. Which element does the root node generally represent?

- A) Constant value
- B) Identifier name
- C) Entire expression or statement
- D) Memory address

**Option: C**

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36. A compiler reports an error because a variable is used outside its declared scope. Which compiler phase detects this issue?

- A) Syntax Analysis
- B) Semantic Analysis
- C) Code Optimization
- D) Linking

**Option: B**

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37. A compiler automatically converts an integer operand into a floating-point operand before arithmetic evaluation. What type of conversion is this?

- A) Narrowing conversion
- B) Implicit widening conversion
- C) Explicit casting
- D) Recursive conversion

**Option: B**

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38. During attribute evaluation, semantic rules are processed from left to right so that inherited information becomes available when required. Which attribute grammar follows this restriction?

- A) Circular grammar
- B) L-attributed grammar
- C) S-attributed grammar
- D) Ambiguous grammar

**Option: B**

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39. A compiler uses shift-reduce actions and reduction tables to parse complex programming constructs. Which parser is most likely being used?

- A) Predictive parser
- B) Recursive descent parser
- C) LR parser

D) LL(1) parser

**Option: C**

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40. A compiler engineer associates semantic computations directly with grammar rules for translation purposes. To what are these semantic rules attached?

- A) Tokens only
- B) Grammar productions
- C) Machine registers
- D) Memory blocks

**Option: B**

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41. During semantic evaluation, a node computes its value entirely using values obtained from child nodes. Which attribute type is involved?

- A) Inherited attribute
- B) Dynamic attribute
- C) Synthesized attribute
- D) External attribute

**Option: C**

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42. A compiler designer prefers a compact internal representation that removes unnecessary non-terminal symbols from derivations. Which representation is best suited?

- A) Parse Tree
- B) Syntax Tree
- C) NFA
- D) Transition Graph

**Option: B**

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43. A program contains the statement `total = amount + tax;` where `tax` has never been declared. Which compiler phase identifies this error?

- A) Lexical Analysis
- B) Parsing
- C) Semantic Analysis
- D) Code Generation

**Option: C**

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44. A compiler automatically changes the datatype of an operand during expression evaluation to avoid type mismatch. What is this process called?

- A) Coercion
- B) Parsing
- C) Reduction
- D) Folding

**Option: A**

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45. Two structures are treated as equivalent because they contain identical field arrangements and datatypes. Which equivalence principle is being applied?

- A) Name equivalence
- B) Scope equivalence
- C) Structural equivalence
- D) Dynamic equivalence

**Option: C**

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46. A compiler implementing LR parsing evaluates semantic actions during reduce operations. Which evaluation strategy aligns best with this parser behavior?

- A) Predictive evaluation
- B) Recursive evaluation
- C) Bottom-up evaluation
- D) Top-down evaluation

**Option: C**

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47. A compiler developer chooses an attribute grammar that can be evaluated naturally during reduce actions without inherited information. Which grammar type is most appropriate?

- A) L-attributed grammar
- B) S-attributed grammar
- C) Ambiguous grammar
- D) Circular grammar

**Option: B**

---

48. In a syntax tree for arithmetic expressions, the terminal values such as variables and constants appear at which location?

- A) Internal nodes
- B) Root node
- C) Leaf nodes
- D) Header nodes

**Option: C**

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49. A programmer explicitly converts a floating-point result into an integer before storing it in a variable. Which operation is being applied?

- A) Promotion
- B) Casting
- C) Translation
- D) Coercion

**Option: B**

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50. After successful syntax checking, a compiler begins verifying meanings, datatypes, and scope rules. Which phase starts next?

- A) Lexical Analysis
- B) Semantic Analysis
- C) Linking
- D) Loading

**Option: B**

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51. A programming language verifies datatype compatibility during compilation and sometimes again during execution. What does this indicate about type checking?

- A) It occurs only during loading
- B) It occurs only during runtime
- C) It can occur during compile time and runtime
- D) It is unnecessary in modern languages

**Option: C**

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52. A parser constructs derivations beginning from tokens and gradually reduces them until the start symbol is obtained. Which parsing approach is demonstrated?

- A) Top-down parsing
- B) Predictive parsing
- C) Bottom-up parsing
- D) Recursive parsing

**Option: C**

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53. A compiler internally represents expressions using parent-child relationships between operators and operands instead of linear text order. Which representation style is being followed?

- A) Sequential representation
- B) Hierarchical representation
- C) Tabular representation
- D) Circular representation



**Option: B**

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54. A compiler phase determines whether expressions, assignments, and declarations carry valid meaning according to language rules. Which aspect is primarily analyzed?

- A) Syntax structure
- B) Meaning of constructs
- C) Register allocation
- D) Instruction scheduling

**Option: B**

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55. Two datatype declarations are accepted as compatible because the language rules consider them logically interchangeable. What does this describe?

- A) Operator precedence
- B) Variable scope
- C) Type equivalence
- D) Memory allocation

**Option: C**

**Answers:**

1. B) Associating semantic rules with grammar productions
2. C) Syntax Tree
3. B) Synthesized attribute evaluation
4. C) S-attributed definition
5. C) L-attributed definition
6. C) Postorder traversal
7. B) Hierarchical program structure
8. C) Inherited attribute
9. B) Type system
10. C) Implicit type conversion
11. C) Semantic Analysis
12. B) Syntax Tree
13. B) Symbol Table
14. C) S-attributed grammar
15. B) Top-down translation
16. C) Stack
17. B) Type checking

18. B) Integer
19. B) Structural equivalence
20. C) Bottom-up evaluation
21. C) Syntax Tree
22. B) L-attributed grammar
23. B) Intermediate Code Generation
24. C) Casting
25. C) Lexical Analysis
26. B) Represents grammar symbols
27. C) Leaf nodes
28. B) Semantic checker
29. C) Name equivalence
30. C) Postorder traversal
31. A) Syntax-directed translation scheme
32. C) Bottom-up parsing
33. A) Type system
34. B) Derived datatype
35. C) Entire expression or statement
36. B) Semantic Analysis
37. B) Implicit widening conversion
38. B) L-attributed grammar
39. C) LR parser
40. B) Grammar productions
41. C) Synthesized attribute
42. B) Syntax Tree
43. C) Semantic Analysis
44. A) Coercion
45. C) Structural equivalence
46. C) Bottom-up evaluation
47. B) S-attributed grammar
48. C) Leaf nodes
49. B) Casting
50. B) Semantic Analysis
51. C) It can occur during compile time and runtime
52. C) Bottom-up parsing
53. B) Hierarchical representation
54. B) Meaning of constructs
55. C) Type equivalence

#### RUBRICS FOR CALCULATION

| Criteria                  | Weightage | Level 4 – Exemplary (90–100 Marks) | Level 3 – Proficient (75–89 Marks) | Level 2 – Developing (50–74 Marks) | Level 1 – Beginning (0–49 Marks) |
|---------------------------|-----------|------------------------------------|------------------------------------|------------------------------------|----------------------------------|
| Number of Correct Answers | 100%      | 90 to 100 correct answers          | 75 to 89 correct answers           | 50 to 74 correct answers           | 0 to 49 correct answers          |

| Criteria                     | Weight<br>age<br>(%) | Level 4 –<br>Exemplary<br>(5 Marks)                                           | Level 3 –<br>Proficient<br>(4 Marks)            | Level 2 –<br>Developing<br>(2–3 Marks)                     | Level 1 –<br>Beginning<br>(0–1 Mark)      |
|------------------------------|----------------------|-------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------|-------------------------------------------|
| Conceptual<br>Understandin g | 30                   | Demonstrate s<br>clear and<br>complete<br>understandin g<br>of the<br>concept | Good<br>understandi ng<br>with minor<br>gaps    | Basic<br>understandi ng<br>with some<br>misconcepci<br>ons | Poor or<br>incorrect<br>understan<br>ding |
| Accuracy of<br>Answer        | 25                   | Completely<br>correct answer<br>with<br>no errors                             | Mostly correct<br>with minor<br>errors          | Partially correct<br>with noticeable<br>errors             | Incorrect or<br>irrelevant<br>answer      |
| Application of<br>Knowledge  | 20                   | Correctly applies<br>concepts to solve<br>the question/prob<br>lem            | Applies<br>concepts<br>with minor<br>mistakes   | Limited or<br>partially correct<br>application             | No or<br>incorrect<br>application         |
| Completeness<br>of Response  | 15                   | Covers all<br>required parts of<br>the<br>question                            | Covers most<br>parts with<br>minor<br>omissions | Covers only<br>some parts                                  | Incomplete or<br>missing<br>response      |
| Clarity &<br>Presentation    | 10                   | Clear, concise,<br>and                                                        | Understand<br>able with                         | Somewhat<br>unclear or                                     | Difficult to<br>understan<br>d            |
| Criteria                     | Weight<br>age<br>(%) | Level 4 –<br>Exemplary<br>(5 Marks)                                           | Level 3 –<br>Proficient<br>(4 Marks)            | Level 2 –<br>Developing<br>(2–3 Marks)                     | Level 1 –<br>Beginning<br>(0–1 Mark)      |
|                              |                      | well-organize<br>d answer                                                     | minor<br>issues                                 | poorly<br>structured                                       |                                           |